

The local, humoral and nervous regulation of the circulation

The local regulation of the circulation

Local blood flow control can be divided into two phases:

Acute control

- ❖ an increase in tissue metabolism increases tissue blood flow
- ❖ **reactive hyperemia**: occurs after the tissue blood supply is blocked for a short time.
- ❖ **active hyperemia** : Occurs When Tissue Metabolic Rate Increases
- ❖ reduced oxygen availability increases tissue blood flow
- vasodilator theory and the oxygen demand theory

Vasodilator theory

- The greater the rate of metabolism or the less the availability of oxygen or some other nutrients to a tissue, the greater the rate of formation of vasodilator substances in the tissue cells.
- ❖ adenosine,
- ❖ carbon dioxide,
- ❖ adenosine phosphate compounds,
- ❖ histamine,
- ❖ potassium ions, and
- ❖ hydrogen ions

Oxygen demand theory

Oxygen causes vascular muscle contraction

Autoregulation of Blood Flow During Changes in Arterial Pressure

Metabolic Mechanisms :

- ❖ when the arterial pressure becomes too great, the excess flow provides too much oxygen and too many other nutrients to the tissues
- ❖ washes out the vasodilators released by the tissues.
- ❖ These nutrients (especially oxygen) and decreased tissue levels of vasodilators then cause the blood vessels to constrict and return flow to nearly normal despite the increased pressure.

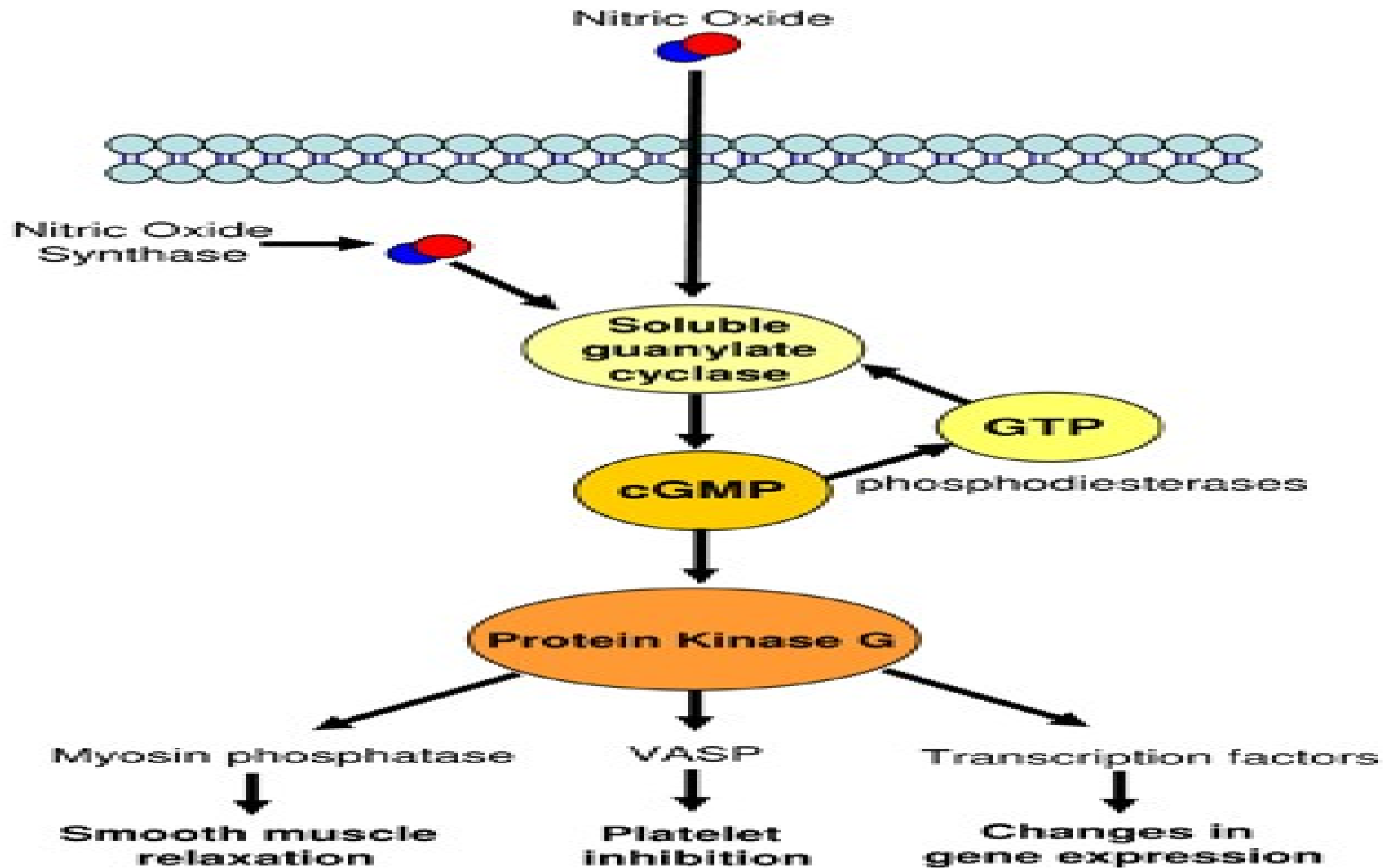
Myogenic Mechanisms:

- high arterial pressure stretches the vessel, this in turn causes reactive vascular constriction that reduces blood flow.

Control of Tissue Blood Flow by Endothelial-Derived Relaxing or Constricting Factors

Nitric Oxide

Endothelin 1



Long-term control

- Arterioles and capillary vessels usually increase both in number and size

Changes in Tissue Vascularity: angiogenesis

- vascular endothelial growth factor (VEGF),
- fibroblast growth factor,
- platelet-derived growth factor (PDGF), and
- Angiogenin

Collateral Circulation

Vascular Remodeling

The humoral regulation of the circulation

VASOCONSTRICTOR AGENTS

- ✓ Norepinephrine and epinephrine
- ✓ Angiotensin II.
- ✓ Vasopressin

VASODILATOR AGENTS

- ✓ Bradykinin
- ✓ Histamine

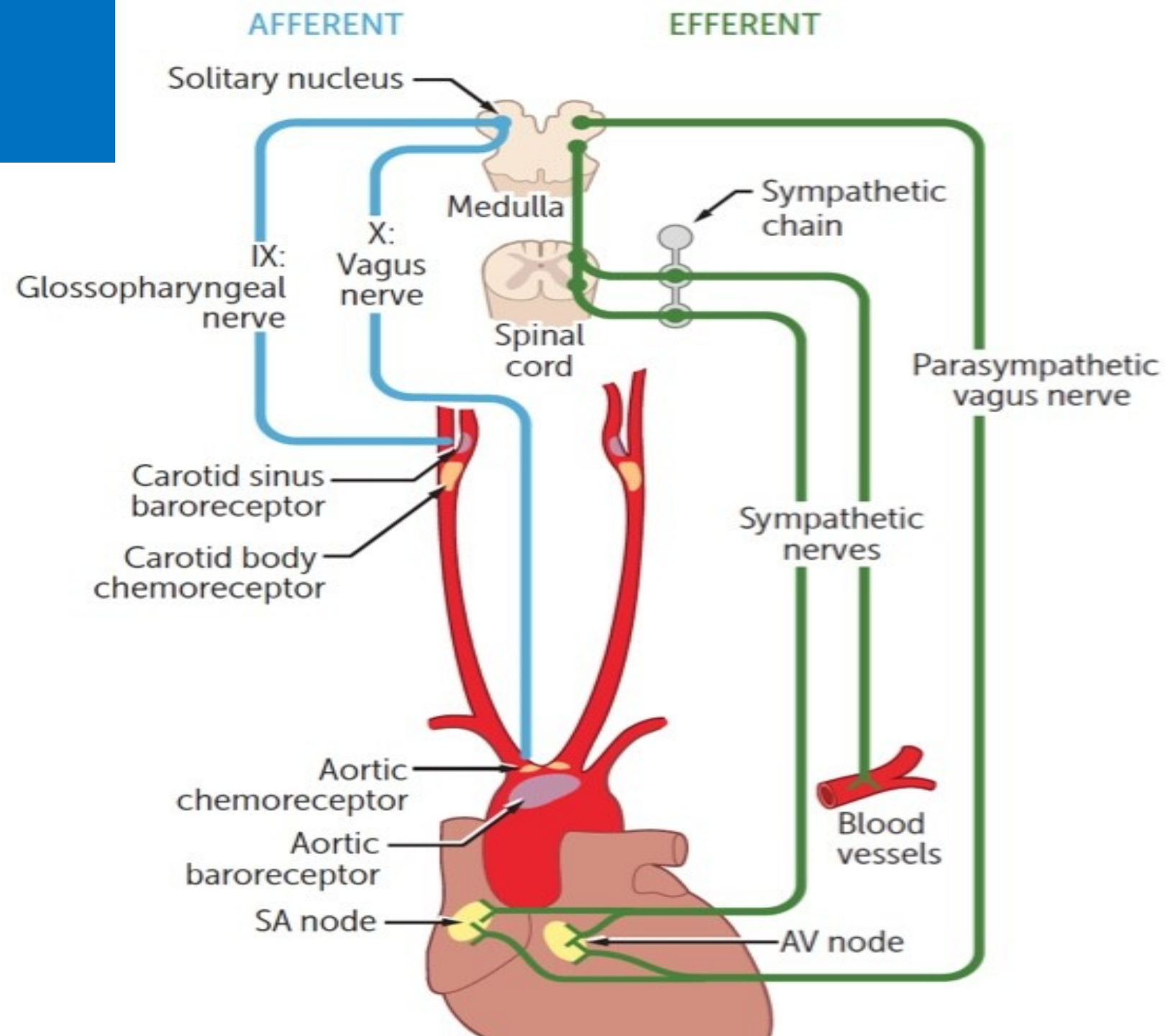
The nervous regulation of the circulation

By autonomic nervous system

Regulation of Arterial blood Pressure

- ❖ Arterial pressure, refers to the pressure measured within large arteries in the systemic circulation.
- ❖ Systolic pressure refers to the maximum pressure within the large arteries when the heart muscle contracts to propel blood through the body.
- ❖ Diastolic pressure describes the lowest pressure within the large arteries during heart muscle relaxation between beating.

Baroreceptor and chemoreceptor Reflex



There are several mechanisms through which the body regulates arterial pressure.

Baroreceptor Reflex

Two forms of baroreceptors.

High-Pressure Baroreceptors

- The carotid baroreceptor responds to both increases and decreases in blood pressure and sends afferent signals via the glossopharyngeal nerve (CN IX).
- The aortic arch baroreceptor responds only to increases in blood pressure, sending its signals through the vagus nerve (CN X).

Low-Pressure Baroreceptors

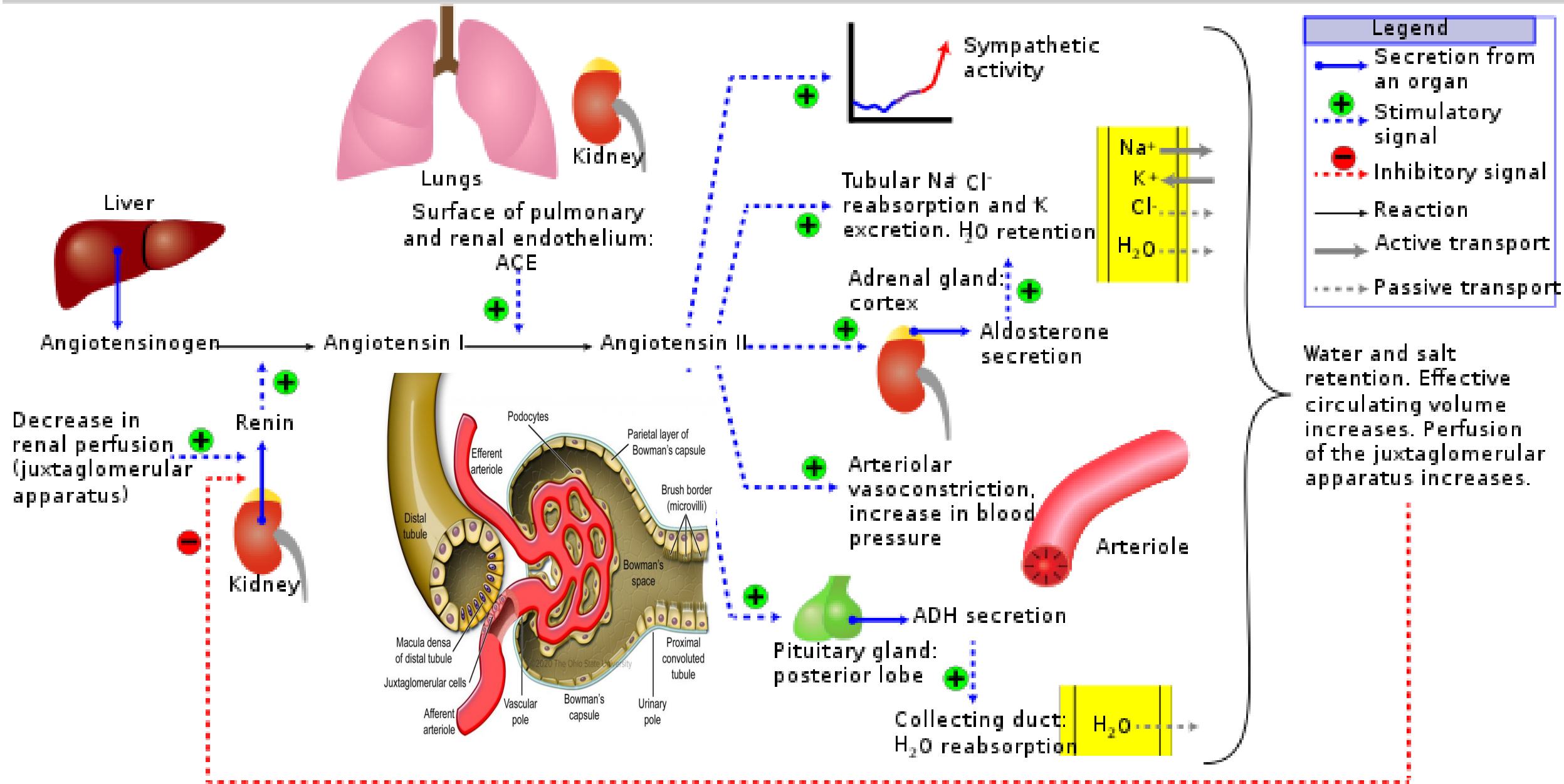
These baroreceptors are present within the low-pressure venous system.

They exist within large veins, pulmonary vessels, and within the walls of the right atrium and ventricle.

Changes in volume largely influence the baroreceptors in the venous system.

Decreased frequency in action potentials in low-pressure scenarios leads to the secretion of antidiuretic hormone, renin, and aldosterone.

Renin-angiotensin-aldosterone system



Antidiuretic Hormone

Increase free water reabsorption in the collecting duct of the nephrons within the kidney

Renin-Angiotensin-Aldosterone System (RAAS)

Angiotensin II has many functions to increase arterial pressure:

- Potent vasoconstriction of arterioles
- Vasoconstriction of the efferent arterioles within the glomerulus of the kidney
- Increased sodium reabsorption within the kidney tubules – causes an increase in blood volume and arterial pressure
- The release of antidiuretic hormone (ADH)
- The release of aldosterone-to increase reabsorption of sodium and secretion of potassium
- The increase in sodium reabsorption leads to passive reabsorption of water and an increase in blood pressure.

Cardiac Output

- ❖ Cardiac output is the quantity of blood pumped into the aorta each minute by the heart.
- ❖ This is also the quantity of blood that flows through the circulation.
- ❖ Venous return is the quantity of blood flowing from the veins into the right atrium each minute.
- ❖ The venous return and the cardiac output must equal each other.

Regulation Of Cardiac Output

- cardiac output is altered by modulating both heart rate (HR) and stroke volume (SV).
- As a result, the regulation of cardiac output is subject to a complex mechanism involving
 - ✓ the autonomic nervous system,
 - ✓ endocrine, and
 - ✓ paracrine signaling pathways

- ❖ Cardiac output is the product of heart rate (HR) and stroke volume (SV) and is measured in liters per minute.
- ❖ HR is the number of times the heart beats in one minute.
- ❖ SV is the volume of blood ejected during ventricular contraction.
- ❖ Not all of the blood that fills the heart by the end of diastole (end-diastolic volume or EDV) can be ejected from the heart during systole.
- ❖ Thus the volume left in the heart at the end of systole is the end-systolic volume (ESV).
- ❖ Thus, the stroke volume is $EDV - ESV$.
- ❖ HR and SV are simultaneously affected by several factors.
- ❖ Cardiac output in humans is generally 5-6 L/min at-rest to more than 35 L/min for athletes during exercise.

- ❖ HR is determined by signals from the sinoatrial node, which automatically depolarizes at an intrinsic rate of 60 to 100 times each minute.
- ❖ SV is the other major determinant of cardiac output and is also affected by several factors.
- ❖ The amount of blood ejected each beat depends on preload, contractility, and afterload.
- ❖ Preload represents all of the factors that contribute to passive muscle tension in the muscles at rest.
- ❖ Preload is proportional to the end-diastolic ventricular volume, or the amount of blood in the ventricles immediately before systole.
- ❖ Greater end-diastolic volumes of blood returned to the heart, increase the passive stretching of the heart muscles.
- ❖ This in turn results in the ventricles contracting with more force- a phenomenon called the Frank-Starling law of the heart.
- ❖ Contractility describes the force of myocyte contraction, also referred to as inotropy.
- ❖ As the force of contraction increases, the heart is able to push more blood out of the heart, and thus increases the stroke volume.
- ❖ The final determinant of stroke volume is afterload.
- ❖ Afterload represents all the factors that contribute to total tension during isotonic contraction.

- ❖ As such afterload can be related to the amount of systemic resistance the ventricles must overcome to eject blood into the vasculature.
- ❖ Afterload is proportionate to systemic blood pressures and is inversely related to stroke volume, unlike preload and contractility.
- ❖ Cardiac output can be increased by a variety of signaling methods including enhancement of sympathetic tone, catecholamine secretion, and circulation of thyroid hormone.
- ❖ These mechanisms increase HR by exerting positive effects via chronotropy (timing), dromotropy (conduction speed), and lusitropy (myocardial relaxation rate).
- ❖ These influences also increase preload through increased venous return via receptor-mediated vasoconstriction.
- ❖ Additionally, contractility is improved through the Frank-Starling mechanism and also by direct catecholamine stimulation.
- ❖ The opposite effects on HR and SV occur when the parasympathetic tone is strengthened in response to decreased oxygen requirements.

Micro circulation

Blood vessels: Classification

1. Elastic vessels (Windkessel vessels):

- Example: Aorta, big arteries
- Pressure storing components
- High ability of recoiling

2. Resistance vessels

Example: small arteries and arterioles

- High muscular component
- Develop high resistance
- Regulate blood flow

3. Exchange vessels: Example, capillaries

- Made up of endothelium and basement membrane.
- Thin enough for exchange

3 types: continuous, discontinuous and fenestrated capillaries

4. Capacitance vessels (big to small veins)

- Very high capacity of distension
- Can accommodate large volume of blood (65% of blood volume)

Blood vessel functions: overview

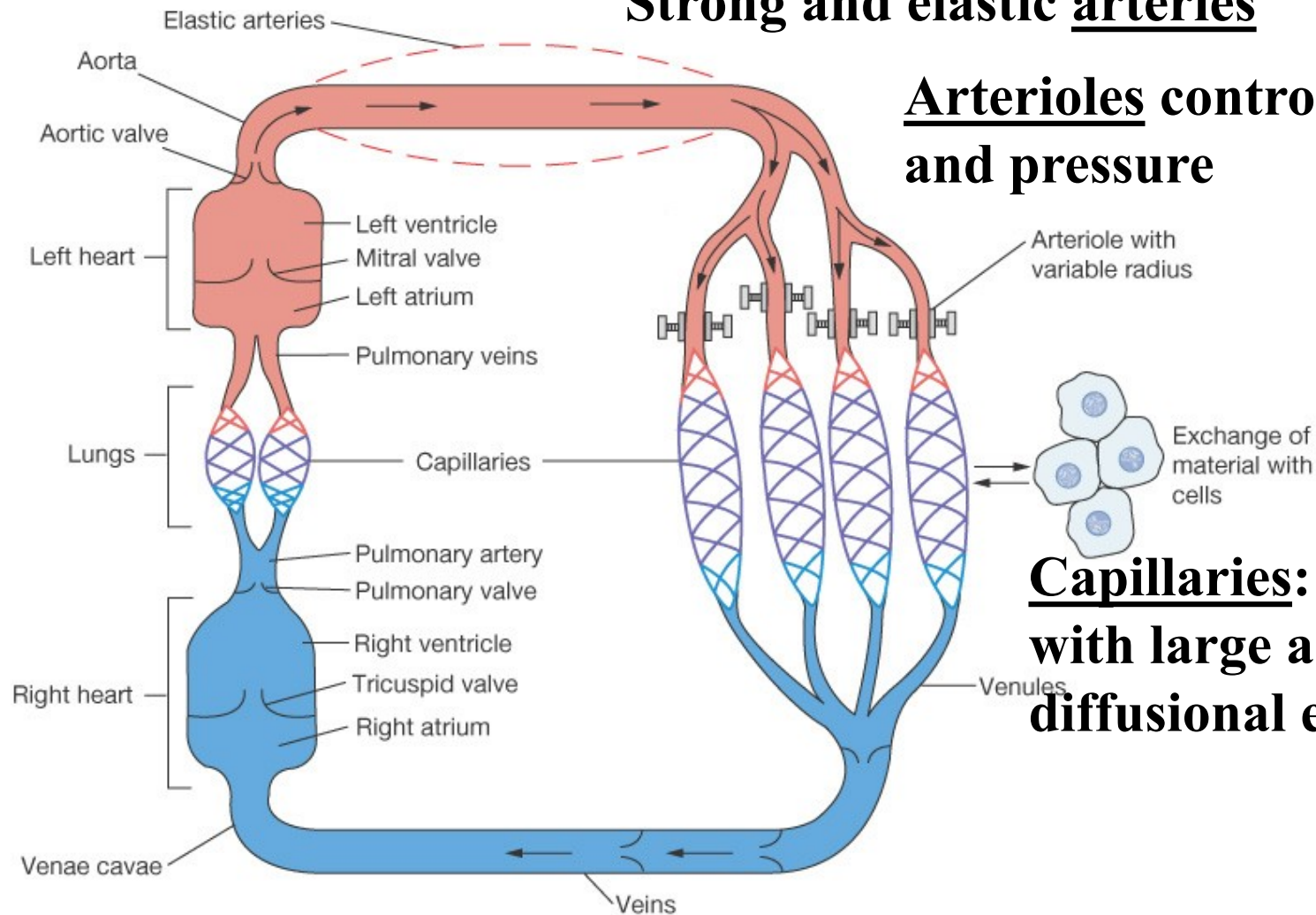
Strong and elastic arteries

Arterioles control blood flow and pressure

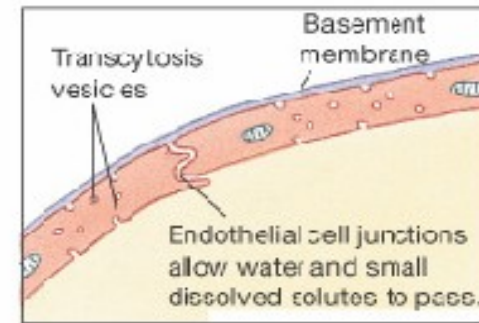
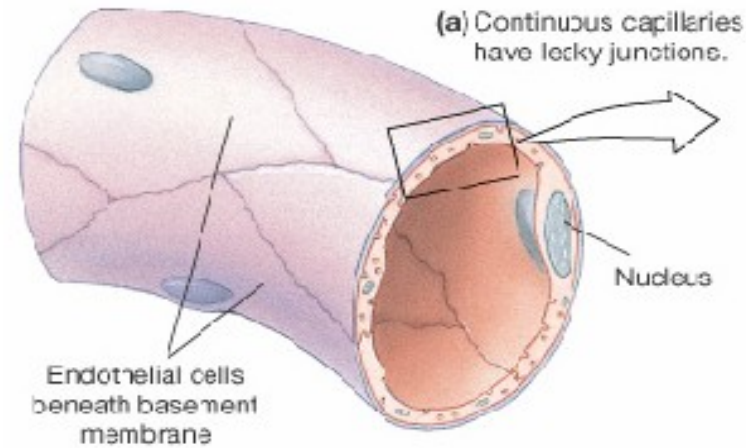
Capillaries: thin and with large area for diffusional exchange

Veins: compliant, large, low resistance

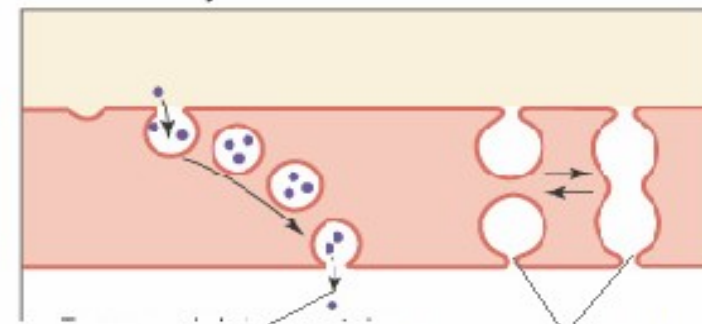
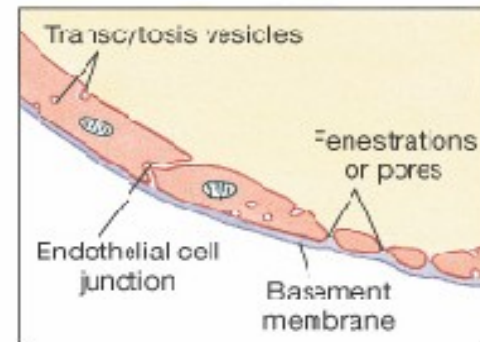
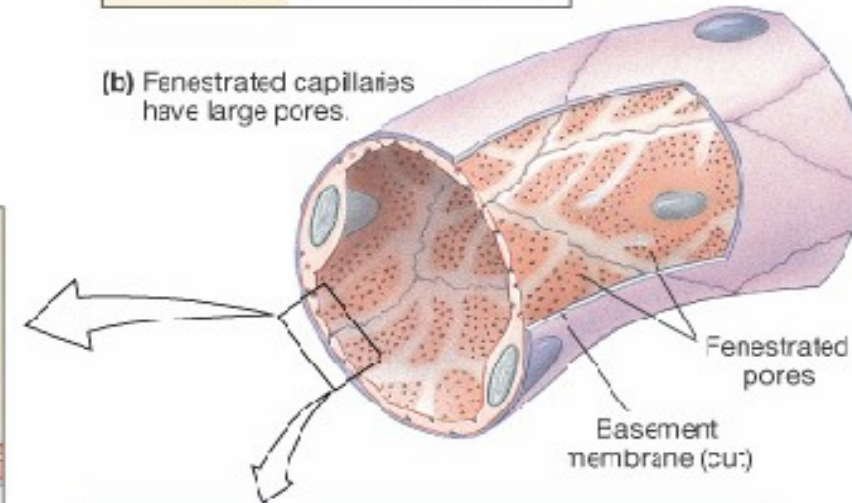
veins have valves & assure blood return to the heart



Types of Capillaries



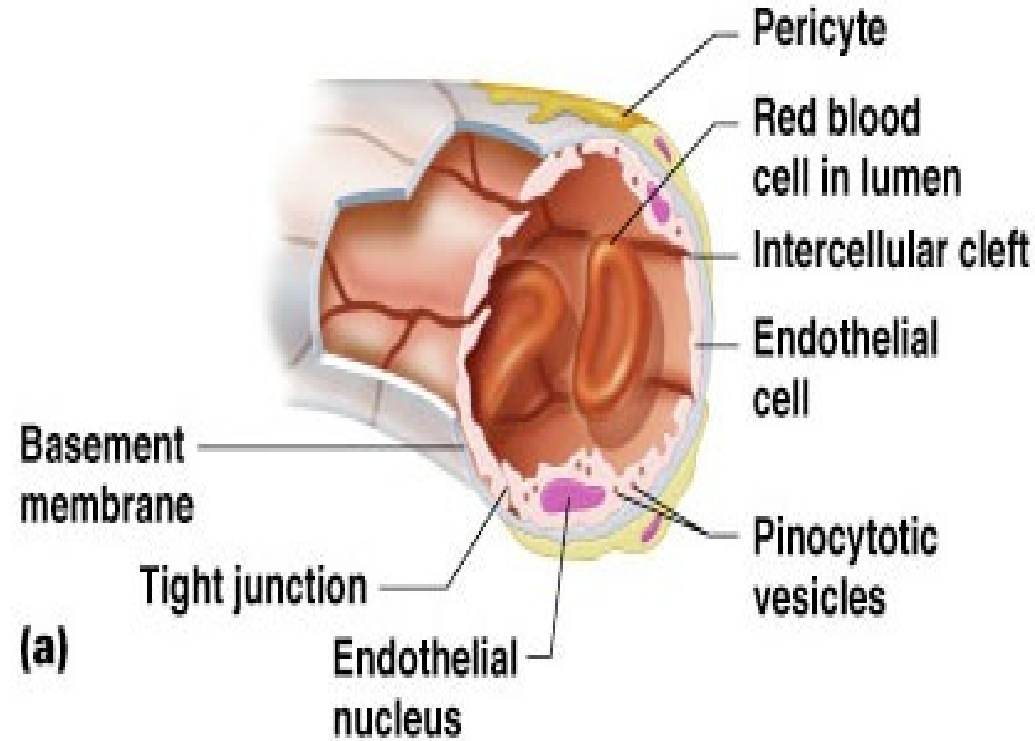
(b) Fenestrated capillaries have large pores.



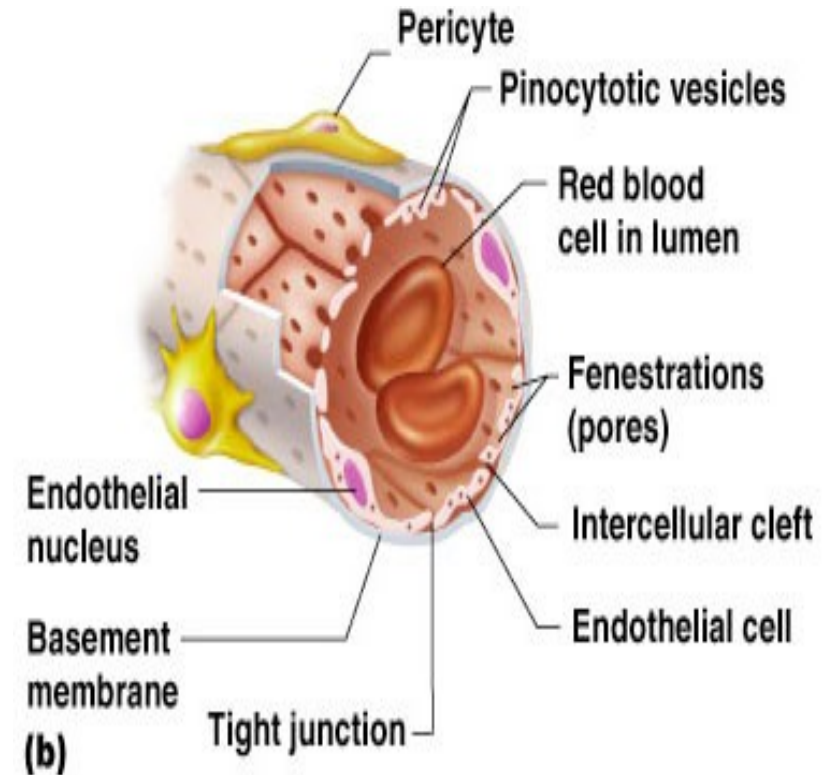
Fenestrated capillaries are found in the glomerulus of the kidney

Transcytosis brings proteins across Endothelium

Some vessels form channels



Continuous Capillaries



Fenestrated capillaries

CAPILLARIES

- Pressure inside is 15 to 35 mmHg
- 5% of the blood is in capillaries
- Necessary for exchange of gases, nutrients, and wastes
- flow is slow and continuous

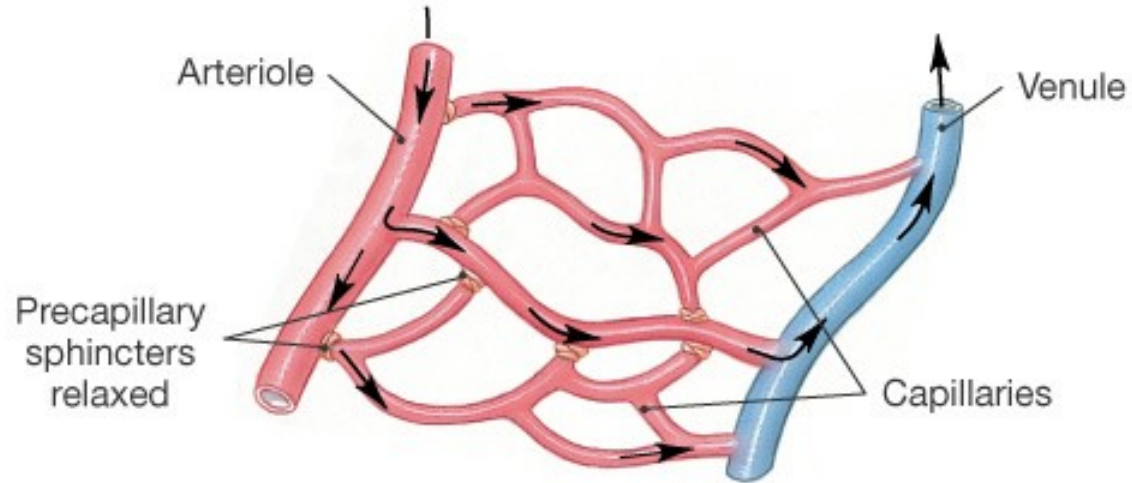
Capillary pressures

- **The capillary pressure (P_c)**, which tends to force fluid outward through the capillary membrane. $P_c = 25 \text{ mm Hg}$
- **The interstitial fluid pressure (P_{if})**, which tends to force fluid inward through the capillary membrane when P_{if} is positive but outward when P_{if} is negative.
- **The capillary plasma colloid osmotic pressure (π_p)**, which tends to cause osmosis of fluid inward through the capillary membrane. $\pi_p = 28 \text{ mm Hg}$
- **The interstitial fluid colloid osmotic pressure (π_{if})**, which tends to cause osmosis of fluid outward through the capillary membrane. $\pi_{if} = 8 \text{ mm Hg}$

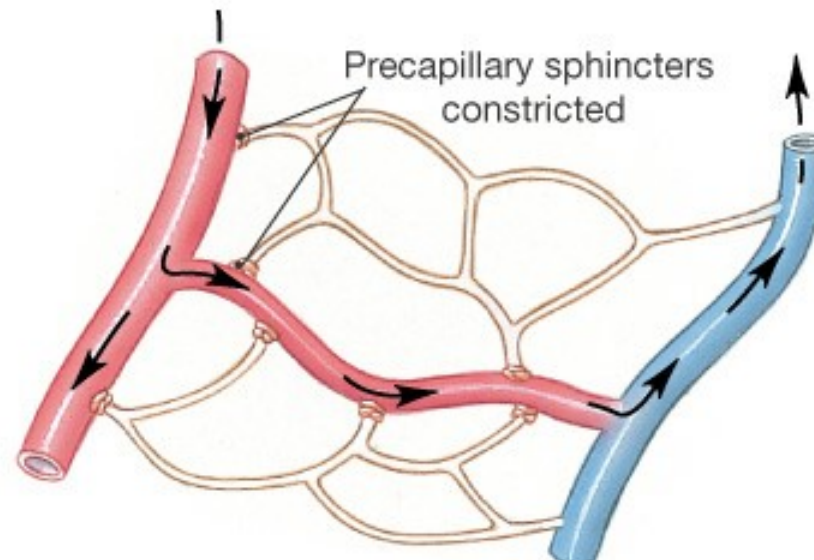
Capillaries

Blood flow
controlled by
pre-capillary
sphincter
muscles *via*
metabolic
regulation
- No innervations

(a) When precapillary sphincters are relaxed, blood flows through all capillaries in the bed.



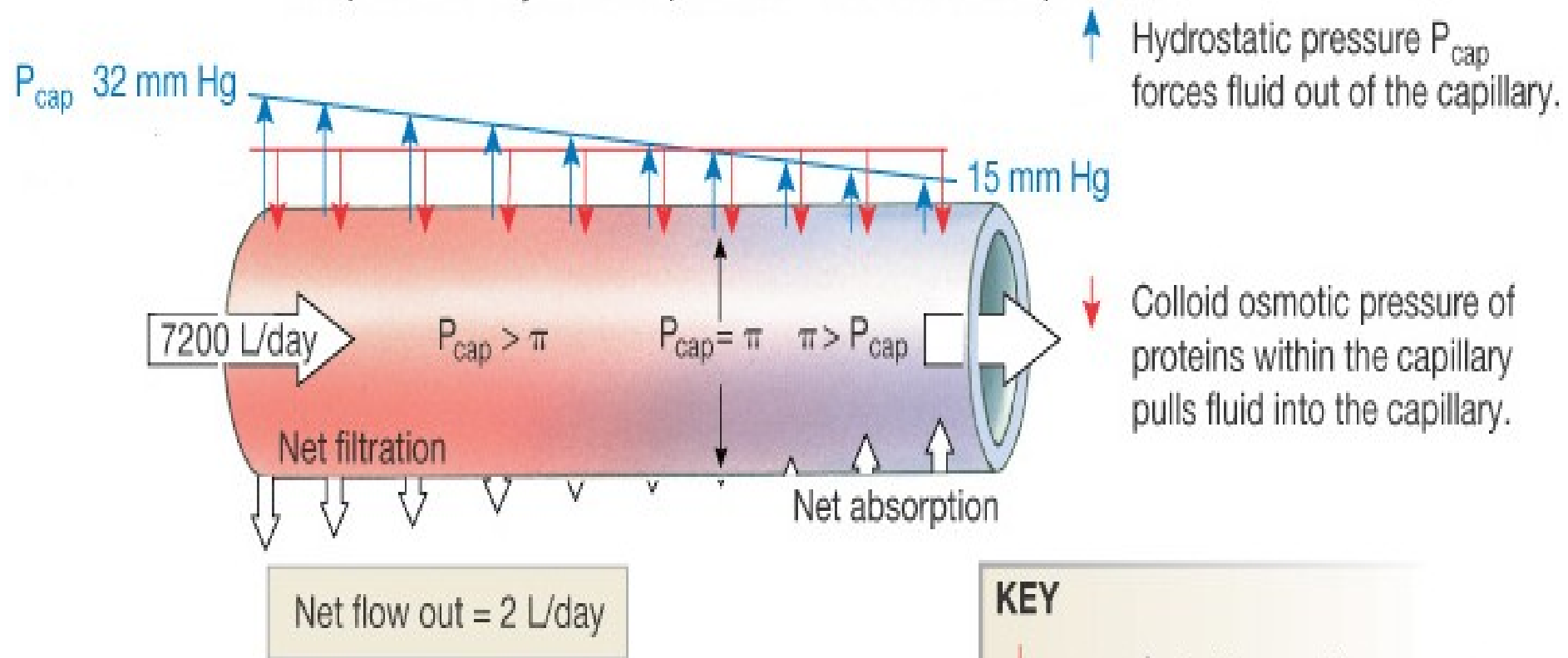
(b) If precapillary sphincters constrict, blood flow bypasses capillaries completely and flows through metarterioles.



Capillary Exchange:

(a) Filtration in systemic capillaries

Net pressure = hydrostatic pressure - colloid osmotic pressure



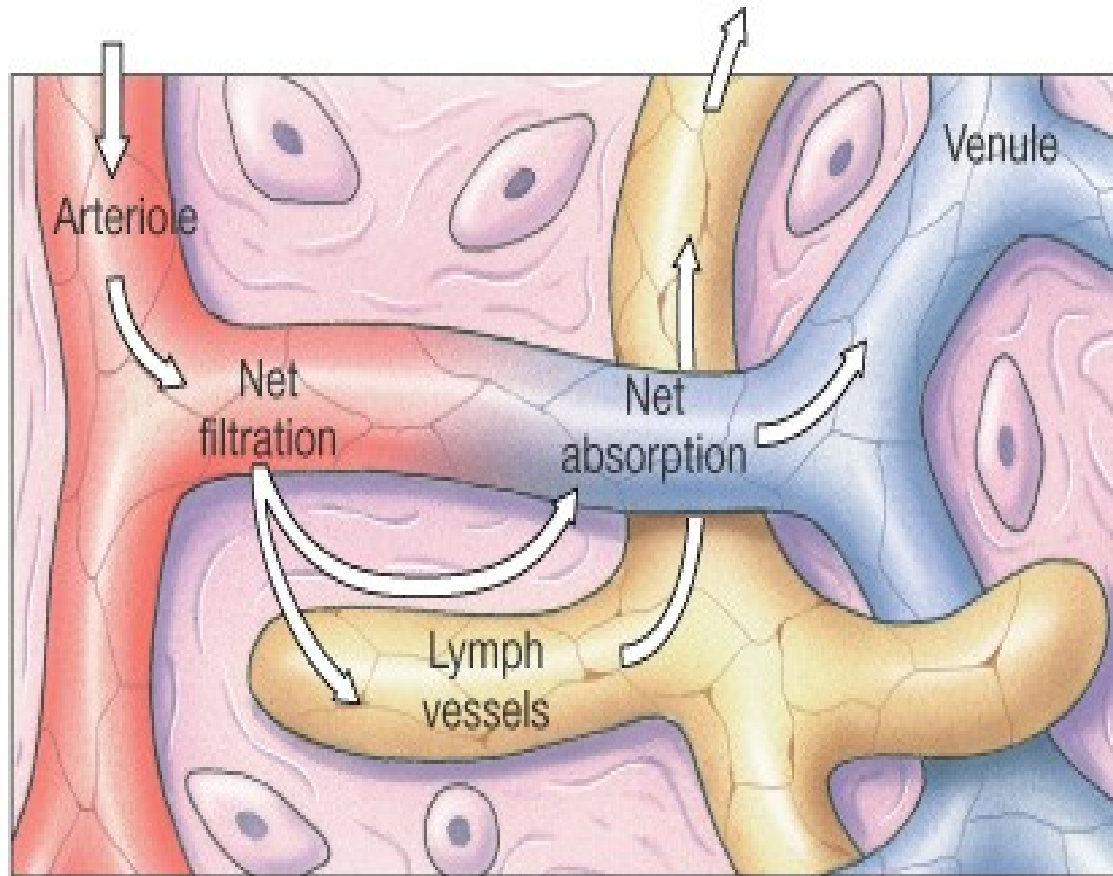
KEY

- ↓ π = Colloid osmotic pressure
- ↑ P_{cap} = Capillary hydrostatic pressure

Net outflow into ECF

- Net filtration – net absorption = net outflow
- About 2 L/day collected by lymph vessels

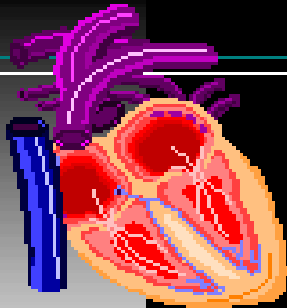
(b) Relationship between capillaries and lymph vessels



The excess water and solutes that filter out of the capillary are picked up by the lymph vessels and returned to the circulation.

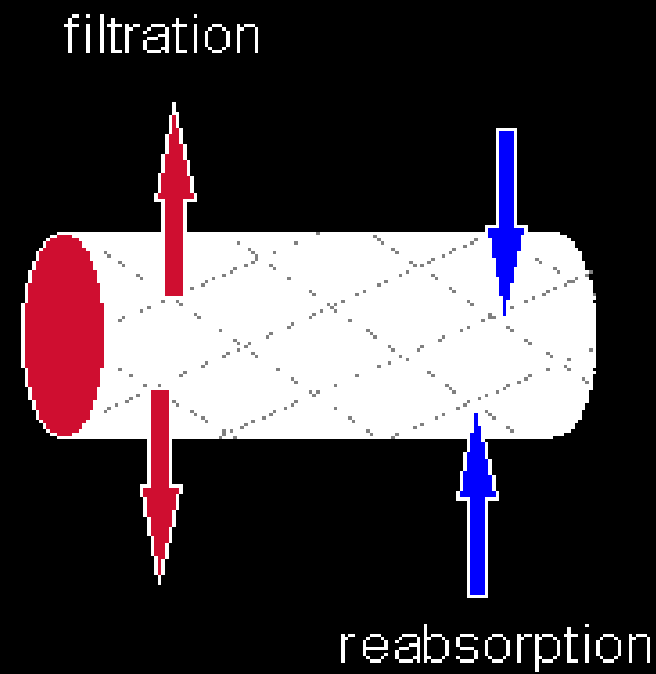
Figure: Fluid exchange at the capillary

Exchange of substances in the capillaries



Capillaries: Bulk Flow

- ◆ Passive movement through **pores**:
 - water & solutes
- ◆ Balances the **extracellular fluid**:
 - plasma
 - interstitial fluid
- ◆ **Filtration** generally exceeds **reabsorption**



HEMODYNAMICS

- Hemodynamics is the study of the relationship between flow, pressure and resistance and other physical principles of blood circulation

Blood flow

- Blood flow (BF) is the amount of blood that moves to a particular organ in a given time
- Total blood flow is equal to CO; 5-6 L/min
- BF is determined by 2 factors
 1. Pressure difference b/n 2 ends of the vessel
i.e., the force that pushes blood through vessels
 2. Resistance of flow, hindrance to flow through vessels

Where **Q = Blood flow**

$$Q = \Delta P / R$$

ΔP = Change in pressure

R = Resistance

Vascular resistance

- **Poiseuille's Law:** Vascular resistance to BF is directly proportional to the length of the vessel and viscosity of blood, but inversely proportional to the 4th power of radius of the vessel.

Where, R = Resistance

$$Q = \frac{\Delta P \pi r^4}{8 \eta l}$$

constant

Q=Blood flow

$$R = 8\eta l / \pi r^4$$

l = Length

η = Viscosity

π = Circle

(3.14)

r = Radius

Factors affecting viscosity of blood

- Velocity of BF: $\uparrow$ Velocity = $\downarrow$ Resistance
- Hematocrit: polycythemia = $\uparrow$ Viscosity = $\uparrow$ PR
Anemia = $\downarrow$ Viscosity = $\downarrow$ PR
- Diameter of blood vessels
- Plasma protein concentration

PR=Peripheral resistance

Velocity of blood flow

- Velocity (v) of BF is directly proportional to the pressure difference (ΔP) and diameter of the vessels, but inversely related to the viscosity of blood and length of the blood vessel

$$v = \Delta P r^2 / 8 \eta l$$

- The quantity of BF (Q) through a vessel is equal to the velocity (v) of flow times the cross-sectional area (A).

$$Q = v \pi r^2, \quad Q = \text{rate of BF}$$

- Substituting the value for velocity of BF in the above equation provides an equation called Poiseuille's law.

$$Q = \frac{\Delta P \pi r^4}{8 \eta l}$$

- This equation states that the rate of blood flow is directly proportional to the fourth power of the radius, indicating that vascular diameter greatly determines rate of BF

Blood flow

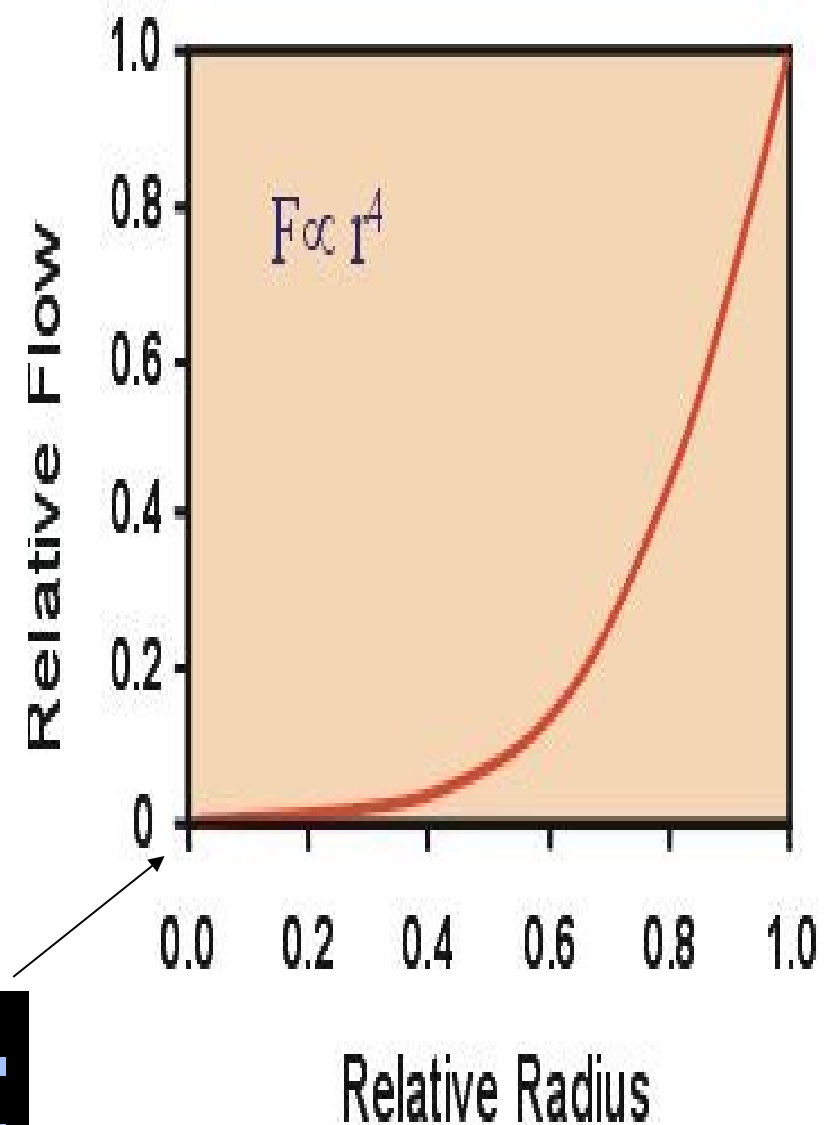
$$BF = \frac{\Delta P}{R}$$

Blood Flow

Blood flow (F) depends on:

- ① Pressure Gradient (ΔP) - *heart*
- ② Resistance (R) - *blood vessels*
 - viscosity
 - vessel length
 - vessel diameter

Vessel diameter is the main determinant of vascular resistance.



Types of Blood Flow

- There are two types of blood flow inside the vessels
 1. Laminar flow
 2. Turbulent flow

Laminar blood flow also called streamline flow

- Blood flows in a **steady rate through long smooth vessels**
- Blood cells move in a **straight line**
- Blood cells at the centre of the vessel move **faster**
- Laminar flow is **silent**

Turbulent flow

- Blood flows in **all directions in a vessel, continuously mixed**
- Turbulent blood flow **produces sound**
- Occurs at blood vessels with **high elastic content and during high velocity**

Turbulent blood flow

- The tendency for turbulent flow increases in direct proportion to the velocity of BF and diameter of blood vessels, but inversely proportional to the viscosity of blood.

$$\text{Re} = \rho D v / \eta$$

Where

Re = Reynold's
number

ρ = density

D = diameter

v = velocity

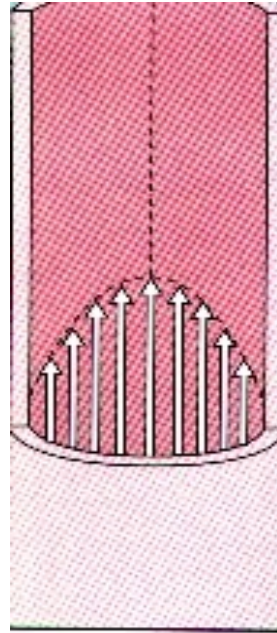
η = viscosity

- Reynold's number is the measure of the tendency for turbulent to occur
- Viscosity of blood is 1/30 Poise, and density of blood is slightly greater than 1.
- When Re rises up from 200 to 400, turbulent flow occurs at some branches
- When Re rises above 2000, turbulent flow occurs even in straight, smooth vessels
- Re is very small in small vessels
- High velocity of BF, large diameter, change in diameter are conditions that favour turbulent flow

LAMINAR vs TURBULENT FLOW

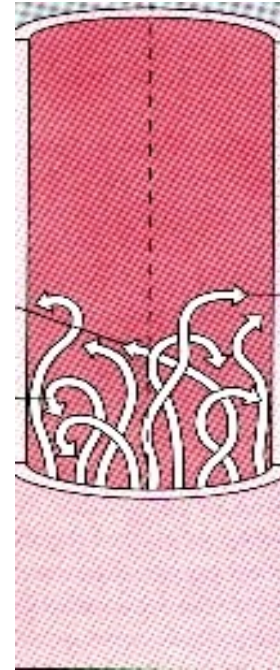
THE REYNOLD'S NUMBER

LAMINAR
FLOW



Parabolic Velocity
Profile
During Laminar
Flow

$$Re = \rho D v / \eta$$



TURBULENT
FLOW

ρ = density
 D = diameter
 v = velocity
 η = viscosity

Resistance to blood flow

- It is the hindrance to blood flow
- Has no direct means of measurement, but can be calculated from values of BF and ΔP in vessels
- If the ΔP is 1 mm Hg and BF 1 ml/sec, then vascular resistance is said to be 1 peripheral resistance unit (PRU)

Total peripheral resistance and total pulmonary resistance

- The rate of BF through the circulatory system at rest is 100 ml/sec or 5 L/min.
- The ΔP between systemic arteries and veins is 100 mm Hg
- Resistance of the entire systemic circulation, called TPR is $100/100 = 1$ PRU.
- TPR ranges between 0.2 and 4 PRU during vascular dilation and constriction

TOTAL PERIPHERAL RESISTANCE

SYSTEMIC CIRCULATION:

RAP=Right atrial
pressure

$$\text{TPR} = \frac{\text{Aortic Pressure} - \text{RAP}}{\text{FLOW}}$$

$$\text{TPR} = \frac{100 - 0 \text{ mmHg}}{83.3 \text{ ml/sec (5 L/min)}} = 1.2 \text{ PRUs}$$

PULMONARY CIRCULATION:

$$\text{Pul. R.} = \frac{\text{Pul. Art. P.} - \text{LAP}}{\text{FLOW}}$$

$$\text{Pul. R.} = \frac{15 - 5 \text{ mmHg}}{83.3 \text{ ml/sec}} = 0.12 \text{ PRUs}$$

Conductance and resistance

- Conductance is a measure of BF, which is given as: $C = \frac{\pi r^4}{8 \eta l}$
- Conductance is the reciprocal of resistance $R = \frac{8 \eta l}{\pi r^4}$
- This indicates that resistance (R) of a vessel is directly proportional to the blood viscosity (η) and length of the vessel, but inversely proportional to the fourth power of the radius (r^4).

Resistance of BF through series vessels

- Assume two vessels are connected in series, having resistance R_1 and R_2 , then the TPR is the sum of the two. $R_T = R_1 + R_2$
- Any amount of resistance in series arrangement must be added.
- Example: $TPR = R_{\text{arteries}} + R_{\text{arterioles}} + R_{\text{capillaries}} + R_{\text{veins}}$

Resistance of vessels in parallel: is equal to the reciprocal of total conductance.

$$C_T = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$